Section I — Data About the Population

One agency, three instruments, and the geography everything else is published on

Democracy's Data

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Democracy is government by the people, and that phrase hides a chore: before the people can govern anything, somebody has to say who and how many they are. Seats in a legislature, votes in the Electoral College, the lines of every district, billions of dollars in federal funds, and rights that switch on at population thresholds — American government hands all of these out **by count of people**. Representation is not a metaphor built on the population. It is arithmetic performed on it. Get the population wrong and the arithmetic distributes power to the wrong places, silently, for a decade at a time.

Knowing the population is also how a country knows itself. Where people live and where they are leaving. How old they are, what they earn, what languages arrive at the door. Whether a city is growing or only seems to be. Every one of those is a fact about the population before it is a fact about politics — and each becomes political the moment somebody divides by it.

Rulers have counted people for as long as there have been rulers, which tells you how much power sits in a count. The Book of Numbers opens with Moses counting men able to bear arms. Rome counted its citizens so thoroughly that its officials, the *censors*, gave us the word — and a census decree is why the Gospel of Luke moves Joseph and Mary to Bethlehem. William the Conqueror's Domesday Book counted England so it could be taxed. In all of these, the count flowed power *upward*: you were counted so that something could be taken from you — a tax, a soldier, an oath. The American Constitution, in its first article, inverted that. It ordered a regular enumeration — the first a nation ever wrote into its founding law — and tied the count to **representation**: you are counted so that power can flow *to* where you live. The count of 1790 was the first census in history taken chiefly to give the counted their share of government.

That is why this section comes first, and why its subject is worth an undergraduate's attention rather than an accountant's. Almost every number you will meet in this book has a population underneath it. A turnout rate is votes divided by people. A claim that a district is majority Black is a claim about a count of people. A stop rate, a registration rate, a share of anything: all of them need somebody to have said how many people are there.

That bottom number, the thing you divide by, is called the **denominator**, and no habit this book teaches will serve you longer than asking where it came from. The top of the fraction usually gets the attention, because the top is the story: the votes, the stops, the dollars. The bottom is where the story quietly goes wrong. Choosing the wrong denominator is the most common way a true number becomes a false sentence, and this book will come back to that constantly.

In the United States, the somebody who says how many people are there is the **Census Bureau**. This section is about data about the population: who produces it, what each product actually measures, and how to read what the Bureau publishes without mistaking one product for another.

The section goes first for a plain reason. **23 of the 95 documents in the rest of this book go to the Bureau for a number.** Not because those documents are about the census. They are about traffic stops, turnout, district maps. They come here because their question needs a denominator, and this is where denominators come from.

01 One agency, three instruments, endlessly confused

The Bureau is not one survey. Its catalogue runs to well over a hundred surveys and censuses, on subjects from housing to the movement of freight. Three of them get almost all of this section, and it is worth saying now why those three and not the others.

It is because they are the three that get mistaken for each other. Each one answers some version of *how many people live here*. Each publishes a figure that looks like a population. Each carries the same agency's name and appears on the same website in the same typeface. Nobody has ever confused the Commodity Flow Survey with a headcount, because nothing about them looks alike. These three look exactly alike, and they are not the same.

The decennial census: a count that allocates power

The **decennial census** is an enumeration — an attempt to write down every person in the country, one by one, rather than estimate them from a sample. It happens once every ten years, and it describes a single moment: where everybody lived on the first of April in a year ending in zero. It is a snapshot, and everything about it follows from that.

It exists because the Constitution orders it. The count decides how many seats in the House of Representatives each state gets, and with them votes in the Electoral College. It then decides where the district lines inside each state can lawfully be drawn. Allocating representation is the job; the statistics are a by-product of it.

The job shapes the instrument. A form that must reach every household in America has to be short, and the 2020 form asked only a handful of questions about each person. So the count is authoritative on *how many*, and nearly silent on almost everything else. It also misses people, and misses them unevenly: the Bureau grades its own count after each census, and one of this section's briefs reads the grade.

The American Community Survey: a rolling sample, with a margin beside every number

The **American Community Survey**, the ACS, is a survey rather than a count. It goes continuously to a sample of about 3.5 million addresses a year, and it asks about everything the census form dropped: income, education, language, rent, veteran status, how you get to work.

Sampling buys three things the once-a-decade count cannot supply. More topics, because a form sent to a sample can afford to be long. Small places, if you let the Bureau pool several years of answers into one estimate; the price of a small-area number is that it describes a five-year blur rather than a moment. And clearer trends, because the ACS is always in the field, so it can show a place changing year by year while the census shows one frame per decade.

The price of sampling is uncertainty, and the ACS states it. Beside every estimate the Bureau prints a **margin of error** — how far off the number could be, and still be doing its job. A census count is published with no margin beside it. An ACS estimate always has one, and reading the estimate while ignoring the margin is this section's second most common mistake.

The population estimates: arithmetic between counts

The **Population Estimates Program**, the PEP, collects nothing from anybody. It takes the last census count and carries it forward with arithmetic: add the births, subtract the deaths, add the people who moved in, subtract the people who moved out. The result is the official population of every state, county and city for each year in which nobody counted.

That makes it the answer to a question the other two instruments cannot touch: how many people live here *this* year. It also makes it exactly as good as the count it started from and the records it adds. An error in the 2020 count does not fade as the decade passes. It is carried forward, faithfully, into every estimate built on top of it.

Ask the three instruments for the population of one county and you will get three different numbers. None of them is wrong. They are answers to three different questions: a count of a moment, a survey's estimate of a period, and arithmetic forward from the last count. Which one you quoted decides what your sentence means.

02 The geography underneath

Everything above is published on a shared substrate, and the substrate has a name: **census geography**. The Bureau divides the country into nested boxes — states, then counties, then tracts, then block groups, then blocks — and every figure it releases is a figure for one of those boxes. There is no census number for a neighborhood as you would draw it. There are numbers for the boxes that overlap it.

This matters far beyond the census, because nearly everybody else borrows the boxes. Election results, health statistics, school data: when they can be joined to a population at all, it is because somebody placed them into census geography. When two datasets refuse to join — a problem that runs through the whole of this book — the usual reason is that they were published on different boxes.

So geography gets a cluster of its own at the end of this section, and it is where the boxes start doing analytic work of their own. Hold the people still and move the boundaries, and the numbers change. One brief there shows a county's racial composition shifting with nothing moving but the lines. Another takes apart the ZIP code, which looks like census geography, is not part of it, and gets asked to stand in for it constantly.

03 A survey that lives in the counting section

One thing here will look misfiled, and the explanation is worth having early. The American Community Survey is a survey: it asks a sample of people questions, which is exactly what Section II is about. It sits in this section anyway, because **answering it is required by law**. The same statute that compels the decennial count compels the ACS, and the envelope says so.

Every survey in Section II is the other kind. You may refuse it, and most people asked do refuse. That difference is not a technicality. It runs through everything the two kinds of data can be made to say, and holding the compulsory survey beside the voluntary ones is one of the more useful comparisons this book will ask you to make.

04 Which population do you mean?

Every section of this book has one mistake it keeps returning to, and this section's is the denominator. Not because the Bureau hides anything — the difficulty is that it publishes so much. For one county in one year there is a count, several ACS windows, and an estimate, each attached to its own moment and its own margin. All of them are in print, and a sentence that says "divided by the population" has not yet said which one it means.

The damage is quiet. Divide a 2024 numerator by a 2020 count and a growing county looks busier than it is. Compare two places using figures from two different instruments and you have compared the instruments, not the places. Quote an ACS share for a small town without its margin and you may be reporting noise with a straight face. No software flags any of this, because every one of these operations runs cleanly.

The briefs ahead make the mistake on purpose, where the stakes are low, so you can watch it happen. One lines up the Bureau's many published answers for a single county and asks which one you mean. Another holds a county's people fixed and moves the boxes they are averaged into. By the end of the section, "what is the denominator, and whose is it" should be a reflex.

05 What is in this section, and how to use it

This book is written in two kinds of document, and this section is the first place you will see the pattern in full. A **chapter** is a reading. It introduces one kind of data: who makes it, under what obligation, what it is published as, and what it structurally cannot say. A **brief** is a lab. It takes data a chapter has introduced, asks one real question of it, and answers it from the file.

The section is organised into 5 clusters. Each cluster is a chapter and then the briefs that use that chapter's data, so the reading always comes before the labs that lean on it.

Cluster	The reading	Briefs
I.1 · The Census and Its Products	What a Census Can and Cannot Do	4
I.2 · The Decennial Census	The Decennial Census	4
I.3 · The American Community Survey	The American Community Survey	3

Cluster	The reading	Briefs
I.4 · Population Estimates	The Population Estimates Program	0
I.5 · Census Geography	Census Geography and Why Data Will Not Join	2

The arc runs from the institution to its instruments to their substrate. First what a census is and what it can and cannot do, then the three instruments in turn, then the geography they are all published on. In full, in reading order:

Cluster	Type	Title	What it is about
I.1	chapter	What a Census Can and Cannot Do	The one thing nothing else can do, and what it cost to keep doing it
I.1	brief	How Many People Live in Allegheny County?	One question, a shelf of published answers, and how to say which one you mean
I.1	brief	Using the Census Data API	Three ways into the Census Bureau, and what the catalog reveals about who the data is for
I.1	brief	Who the Count Missed	The census grades itself, state by state, and mostly returns a blank
I.1	brief	Hispanic Origin Is Not a Race on the Census Form	Two questions about one person on the 2020 census form, and the two boxes people use to answer neither
I.2	chapter	The Decennial Census	Seven questions asked once a decade, and the people it does not find
I.2	brief	Apportionment: How a Count Becomes Seats	Why the Constitution orders a census, the six rules that turn it into seats, and what happens when the margin is smaller than the instrument
I.2	brief	Regional Population Shift and House Seats	Nobody proposed it, nobody voted on it, and no single decade announced it
I.2	brief	Measuring Race at Eight Geographic Scales	What the 2020 census says about race, at eight scales, and why only the smallest one describes anyone
I.2	brief	When a Scatter Plot Has Too Many Points	Every populated census block in Georgia, and the 88% of them you cannot see
I.3	chapter	The American Community Survey	Three windows onto the same country, one of them closed, and a margin beside every number that has one
I.3	brief	Measuring Migration with ACS Flow Estimates	One question on an American survey form, the map it draws, and how much of that map is not there
I.3	brief	Everybody Moved, Almost Nobody Left	A year of interstate migration drawn as a chord diagram, and the aggregation you have to perform before a circle will hold it
I.3	brief	Section 203 Language Coverage Determinations	Section 203 of the Voting Rights Act, and the survey that decides who gets a ballot they can read
I.4	chapter	The Population Estimates Program	The number you get when nobody counted, and the arithmetic that produces it
I.5	chapter	Census Geography and Why Data Will Not Join	Why so much data will not join
I.5	brief	The Modifiable Areal Unit Problem	One county's census blocks, and what moving the boxes does to the share
I.5	brief	What a ZIP Code Is, and What It Is Not	A routing key for mail trucks, and everything it gets asked to stand for

That is 5 chapters and 13 briefs, read from the book's own section map as this page is built. Reorganise the section and rebuild, and the tables above follow.

06 You will meet this data again

Quantity	Docs
Docs outside this section	95
Of those, that go to the Census Bureau themselves	23
Of those, that read this section's own files	0

Two numbers in that table, and the gap between them is the honest shape of this section's importance.

23 documents elsewhere in this book go to the Census Bureau themselves. They come for a turnout denominator, a benchmark for traffic stops, a check on a guess about somebody's race, or a population to weight a survey against. They are not about the census, and they cannot be finished without it.

But **0** of them read this section's own files. It would be easy, and wrong, to claim that everything else in the book rests on Section I. It does not. The true version is narrower and more useful: this is the source the rest of the book keeps going back to, independently, on its own terms. That is exactly why it is worth meeting here, rather than first meeting it as somebody else's denominator.

07 How to read this section

Start with what a census can and cannot do, the chapter that opens the first cluster. Then take the instrument chapters in cluster order: the decennial census, the ACS, the estimates, and finally the geography. They are ordered by how often they are mistaken for each other, not by importance.

The briefs can be picked by interest, and each one names the chapter it leans on. If you only have time for two readings, take the opening chapter and the chapter on the instrument you are actually about to use. If you already know which file you need, the access brief is the fastest route in. Read its finding first, because the Bureau has more than a dozen published answers in print for one county's population, and naming which one you mean is the whole skill.

08 Sources

- The corpus itself. The clusters, their order and each document's title are read from this book's own section map (`_lib/make-index.py`) and from each document's front matter, so this page cannot disagree with the section it opens; reorganise the section and rebuild, and the tables above follow.

- U.S. Constitution, Article I, Section 2; the Census Act, 13 U.S.C., including § 221, which makes answering compulsory; Public Law 94-171 (1975); and Section 203 of the Voting Rights Act. Between them these four obligations produce almost everything in this section.